

Milestone 3

Model Evaluation and Explainability

Objective

The goal of Milestone 3 is to verify, interpret, and validate the performance of the trained scanner classification model.

This stage ensures that the system is not only accurate, but also reliable, unbiased, and explainable, which is essential for real-world deployment in document forensics and scanner source identification.

Model Evaluation Strategy

After training, the model must be evaluated on unseen validation data.

This prevents overfitting and ensures the model has learned *general patterns* rather than memorizing the training images.

The evaluation is performed using the following theoretical measures:

1. Accuracy

Accuracy represents the proportion of total predictions that are correct.

It provides a general idea of model performance but does not reveal class-wise behavior.

2. Precision

Precision measures how many of the predicted samples for a class are actually correct.

It indicates how reliable the model's positive predictions are.

3. Recall

Recall measures how many of the actual samples of a class are correctly identified.

It reflects the model's ability to detect all instances of a scanner type.

4. F1-Score

The F1-score is the harmonic mean of precision and recall.

It is especially important when class distributions are uneven.

Confusion Matrix Analysis

A confusion matrix is used to visualize the performance of the classifier.

It compares the true labels with the predicted labels and shows:

- Correct classifications along the diagonal
- Misclassifications in off-diagonal cells

This matrix reveals which scanner models are being confused with one another and helps identify class-level weaknesses.

Classification Report

A classification report summarizes the performance of each class using:

- Precision
- Recall
- F1-score
- Support (number of samples per class)

This report ensures that the model performs consistently across all scanner categories and is not biased toward any one brand.

Explainability using Grad-CAM

Deep learning models are often considered black boxes.

To make the decision process transparent, Gradient-weighted Class Activation Mapping (Grad-CAM) is used.

Grad-CAM highlights the regions of the input image that most strongly influence the model's prediction.

By overlaying these activation maps on the original image, we can visually confirm that the model focuses on:

- Scanner body structure
- Logo areas
- Button layouts
- Surface textures

rather than irrelevant background regions.

This provides visual evidence that the model learns meaningful features.

Model Validation

The final model is validated by selecting the version that performs best on the validation set.

This ensures:

- Better generalization
- Reduced overfitting
- Stable real-world performance

Only the best-performing model is used for further testing and deployment.

Conclusion

Milestone 3 establishes the trustworthiness of the scanner classification system by combining:

- Quantitative performance metrics
- Visual diagnostic tools
- Explainable AI techniques

This milestone transforms the model from a prediction engine into a verifiable and interpretable intelligent system.